

tf.split

 [View source on GitHub \(https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L1710-L1789\)](https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L1710-L1789)

Splits a tensor value into a list of sub tensors.

```
tf.split(  
    value, num_or_size_splits, axis=0, num=None, name='split'  
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
Distributed training with Core APIs and DTensor (https://www.tensorflow.org/guide/core/distribution) Understanding masking & padding (https://www.tensorflow.org/guide/keras/understanding_masking_and_padding)	Using DTensors with Keras (https://www.tensorflow.org/tutorials/distribute/dtensor_keras_tutorial) Distributed training with DTensors (https://www.tensorflow.org/tutorials/distribute/dtensor_ml_tutorial) Convolutional Variational Autoencoder (https://www.tensorflow.org/tutorials/generative/cvae) MoViNet for streaming action recognition (https://www.tensorflow.org/hub/tutorials/movinet) Bayesian Modeling with Joint Distribution (https://www.tensorflow.org/probability/examples/Modeling_with_JointDistribution)

See also [tf.unstack](https://www.tensorflow.org/api_docs/python/tf/unstack) (https://www.tensorflow.org/api_docs/python/tf/unstack).

If `num_or_size_splits` is an `int`, then it splits `value` along the dimension `axis` into `num_or_size_splits` smaller tensors. This requires that `value.shape[axis]` is divisible by `num_or_size_splits`.

If `num_or_size_splits` is a 1-D Tensor (or list), then `value` is split into `len(num_or_size_splits)` elements. The shape of the `i`-th element has the same size as the `value` except along dimension `axis` where the size is `num_or_size_splits[i]`.

For example:

```
>>> x = tf.Variable(tf.random.uniform([5, 30], -1, 1))  
>>>  
>>> # Split `x` into 3 tensors along dimension 1  
>>> s0, s1, s2 = tf.split(x, num_or_size_splits=3, axis=1)  
>>> tf.shape(s0).numpy()  
array([ 5, 10], dtype=int32)  
>>>  
>>> # Split `x` into 3 tensors with sizes [4, 15, 11] along dimension 1  
>>> split0, split1, split2 = tf.split(x, [4, 15, 11], 1)  
>>> tf.shape(split0).numpy()  
array([5, 4], dtype=int32)  
>>> tf.shape(split1).numpy()  
array([ 5, 15], dtype=int32)  
>>> tf.shape(split2).numpy()  
array([ 5, 11], dtype=int32)
```

Args	
value	The Tensor to split.
num_or_size_splits	Either an <code>int</code> indicating the number of splits along <code>axis</code> or a 1-D integer Tensor or Python list containing the sizes of each output tensor along <code>axis</code> . If an <code>int</code> , then it must evenly divide <code>value.shape[axis]</code> ; otherwise the sum of sizes along the split axis must match that of the <code>value</code> .
axis	An <code>int</code> or scalar <code>int32</code> Tensor. The dimension along which to split. Must be in the range <code>[-rank(value), rank(value))</code> . Defaults to 0.
num	Optional, an <code>int</code> , used to specify the number of outputs when it cannot be inferred from the shape of <code>size_splits</code> .

name	A name for the operation (optional).
Returns	
if num_or_size_splits is an int returns a list of num_or_size_splits Tensor objects; if num_or_size_splits is a 1-D list or 1-D Tensor returns num_or_size_splits.get_shape[0] Tensor objects resulting from splitting value.	
Raises	
ValueError	If num is unspecified and cannot be inferred.
ValueError	If num_or_size_splits is a scalar Tensor.

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